

SimpleNet: A Simple Network for Image Anomaly Detection and Localization

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Addressing Key Challenges in Anomaly Detection



Reconstruction methods can fail to identify minor anomalies.



Synthetic anomalies in image space don't match real-world defects.



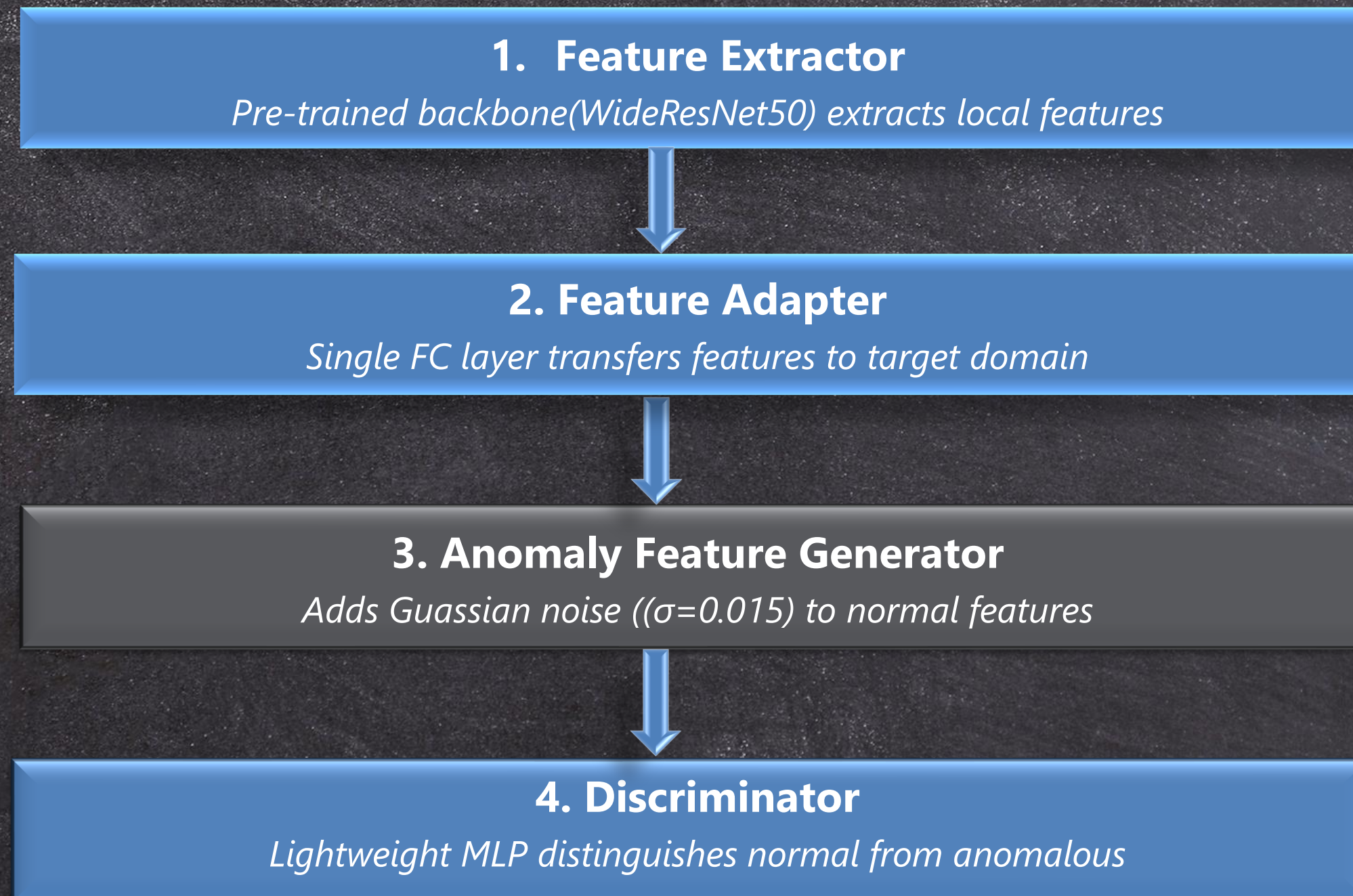
Pretrained models, like those from ImageNet, introduce domain bias.



Embedding-Based (PatchCore, PaDiM) are memory-intensive and slow.

Problem: Cannot collect enough labeled defect examples.

SimpleNet Architecture



*Key Insight: Synthesize anomalies in **feature space**, not image space!*

Why SimpleNet Works ?

- 💡 Feature adaptation reduces domain bias and improves accuracy.
- 📋 Noise in feature space creates more realistic synthetic anomalies.
- 🔑 Lightweight discriminator detects anomalies efficiently and quickly.
- 🌐 One-class novelty detection extends beyond industrial defects.

State-of-the-Art Performance

99.6%

Detection AUROC

↑ 55.5% error reduction vs. PatchCore

98.1%

Localization AUROC

Precise pixel-level defect maps

77 FPS

Inference Speed

8× faster than PatchCore

9/15

Best on MVTec Categories

100% on hazelnut, metal nut, wood

Best accuracy + Fastest speed simultaneously!

Applications & Current Scope

Applications :



Manufacturing Quality Assurance



Real-Time Quality Control Systems



Adaptable Across Product Lines

Current Scope :



Requires Per-Category Training



Dependency on Pre-trained Features



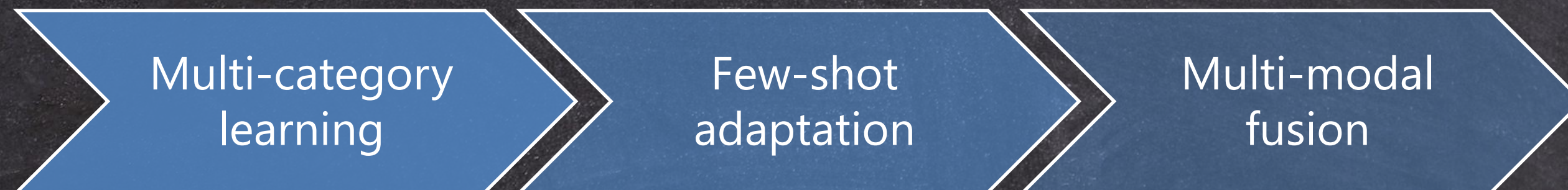
Fixed Noise Generation Strategy

Key Takeaways & Future Exploration

Key Takeaways:

- Feature-space synthesis > Image-space synthesis
- Domain adaptation crucial for pre-trained models
- Simplicity doesn't sacrifice performance
- Ready for industrial deployment TODAY

Future Exploration:



Thank you 😊 !